

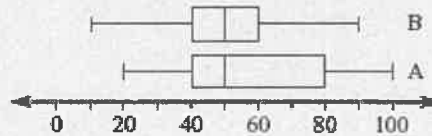
2 For each set of data below complete the following tasks.

- i Find the range.
- ii Find the lower quartile (Q_1) and the upper quartile (Q_3).
- iii Find the interquartile range.
- iv Locate any outliers.
- v Draw a boxplot.

- a 2, 2, 3, 3, 3, 4, 5, 6, 12
- b 11, 12, 15, 15, 17, 18, 20, 21, 24, 27, 28
- c 2.4, 0.7, 2.1, 2.8, 2.3, 2.6, 2.6, 1.9, 3.1, 2.2

3 Compare these parallel boxplots A and B and answer the following as True or False.

- a The range for A is greater than the range for B.
- b The median for A is equal to the median for B.
- c The interquartile range is smaller for B.
- d 75% of the data for A sits below 80.



4 Consider the simple bivariate data set

x	1	4	3	2	1	4	3	2	5	5
y	24	15	16	20	22	11	5	17	6	8

- a Draw a scatter plot for the data.
- b Describe the correlation between x and y as positive or negative.
- c Describe the correlation between x and y as strong or weak.
- d Identify any outliers.

7 Calculate the mean and standard deviation for these small data sets. Round standard deviation to one decimal place.

- a 4, 5, 7, 9, 10
- b 1, 1, 3, 5, 5, 9

1 The number of flying foxes taking refuge in two different fig trees was recorded over a period of 14 days. The data collected is given here.

Tree 1	56	38	47	59	63	43	49	51	60	77	71	48	50	62
Tree 2	73	50	36	82	15	24	73	57	65	86	51	32	21	39

- a Find the IQR for:
 - i tree 1
 - ii tree 2
- b Identify any outliers for:
 - i tree 1
 - ii tree 2
- c Draw parallel boxplots for the data.
- d By comparing your boxplots, describe the difference in the ways the flying foxes use the two fig trees for taking refuge.



The approximate number of shoppers in an air-conditioned shopping plaza was recorded for 14 days, along with the corresponding maximum daily outside temperatures for those days.



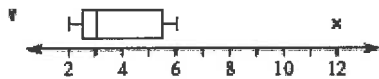
Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Max. daily temp. (T) ($^{\circ}\text{C}$)	27	26	28	33	38	36	28	30	32	25	25	27	29	33
No. of shoppers (N)	1050	950	1200	1550	1750	1800	1200	1450	1350	900	850	700	950	1250

2 Draw a scatter plot for the number of shoppers versus the maximum daily temperatures, with the number of shoppers on the vertical axis, and describe the correlation between the variables as 'positive', 'negative' or 'none'.

- d Use technology to determine the least squares regression line for the data.
- e Use your least squares regression equation to estimate:
 - i the number of shoppers on a day with a maximum daily temperature of 24°C
 - ii the maximum daily temperature if the number of shoppers at the plaza is 1500.

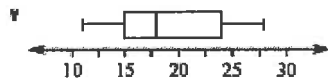
Answers

2 a i 10 ii $Q_1 = 2.5, Q_3 = 5.5$ iii 3 iv 12



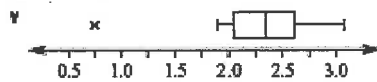
b i 17 ii $Q_1 = 15, Q_3 = 24$

iii 9 iv none

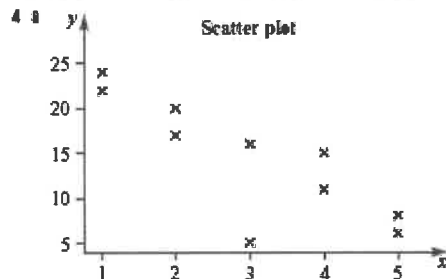


c i 2.4 ii $Q_1 = 2, Q_3 = 2.61$

iii 0.50 iv 0.7



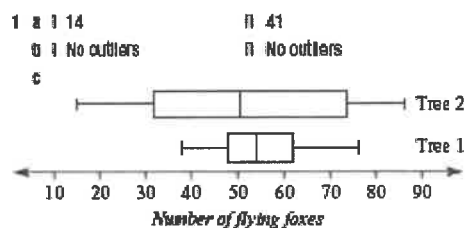
3 a False b True c True d True



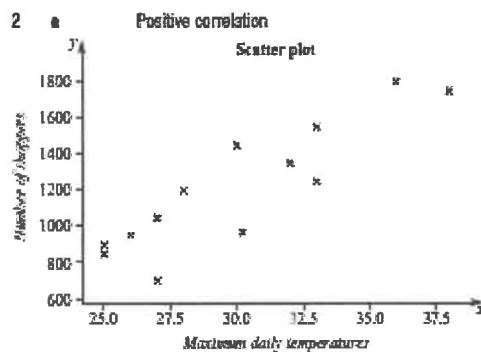
b Negative c Weak d (3, 5)

7 a Mean = 7, $\sigma = 2.3$

b Mean = 4, $\sigma = 2.8$



d More flying foxes regularly take refuge in tree 1 than in tree 2, for which the spread is much greater.



b $y = 92.8571x - 1491.67$

c i 737 ii 32.2°C

d $y = 74.5585x - 1010.06$

e i 779 ii 33.7°C